MIL-A-18850C(OS)
19 Nov 1970
Superseding
MIL-A-18850B
20 March 1963

MILITARY SPECIFICATION

AMPLIFIER, SERVO

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification describes the requirements for the procurement, inspection, and test of a servo amplifier, NAVORD LD 162565 and NAVORD LD 296726. The amplifier is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 5990

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 162565	Servo Amplifier Assembled (Mods 2 and 3)
NAVORD LD 162754	Shipping (Packing) Condition (Mods 2 and 3)
NAVORD LD 296726	Servo Amplifier Assembled (Mod 0)
NAVORD LD 495865	Shipping (Packing) Condition (Mod 0)
NAVORD Drawing 877408	Servo Amplifier Schematic Diagram

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Ordnance Design Requirements; General Miscellaneous

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The servo amplifier is an electronic device which receives signals from error sensing elements and mixes and amplifies those signals in a voltage and dc amplifier, the output of which controls three electromagnetic clutches. When any of the three clutches

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

moves to the desired position, a signal is sent back by electromechanical means to inform the servo amplifier that the desired position of the clutch has been attained and at this point to cancel the error signal.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the servo amplifier shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing servo amplifiers that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the servo amplifier shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 162565 and LD 296726.

3.2.4 Finishes, coatings, construction, and dimensions. The finishes, coatings, construction, and dimensions for the servo amplifier shall be in accordance with applicable drawings, specifications, and publications listed in LD 162565 and LD 296726.

3.2.5 Moisture, fungus, and corrosion. The servo amplifier shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341.

3.2.6 Marking. The marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 162565 and LD 296726.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or the appointed agent for interpretation, clarification, and resolution (or correction).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

3.3 Performance and product characteristics. The servo amplifier shall have the product characteristics and meet the performance requirements given below:

3.3.1 Signal test voltages for Sections 3.3.4, 3.3.6, 3.3.8, 3.3.10, 3.3.12, 4.4.4.3.4, 4.4.4.3.5, 4.4.4.3.6, 4.4.4.3.7, and 4.4.4.3.8, when applied to terminals C and F, 14 and F, 15 and F, and 16 and F on Connector 404, NAVORD Drawing 877408, shall be of the same phase as the power source on terminals 20 and 19, respectively; that is, terminals C, 14, 15, and 16 shall have the same phase relationship as terminal 20.

3.3.1.1 Signal test voltages for Sections 3.3.5, 3.3.7, 3.3.9, 3.3.11, 3.3.13, 4.4.4.3.4(a), 4.4.4.3.5(a), 4.4.4.3.6(a), 4.4.4.3.7(a), and 4.4.4.3.8(a), when applied to terminal C and F, 14 and F, 15 and F, and 16 and F, shall have the same phase relationship as the power source on terminals 19 and 20, respectively; that is, terminals C, 14, 15, and 16 shall have the same phase relationship as terminal 19.

3.3.2 Unless otherwise specified, all references to terminals shall apply to Connector 404, NAVORD Drawing 877408.

3.3.3 Load currents. The load currents between terminal E (150 volts) and terminals H, J, K, L, M, and N shall be balanced at 6.5 plus or minus 1.5 milliamperes (ma). This balance is obtained by adjusting the potentiometers as specified in 4.4.4.2.

3.3.3.1 When the output currents are balanced as specified in 4.4.4.2, a rotation of 1 degree (maximum) in the follow-up potentiometer indicators specified in 4.4.3.3.1, shall give 13 plus 1 ma minus 1.5 ma in one output coil and 1 plus or minus 1 ma in the other coil. This condition shall prevail on all three channels: port, starboard, and rudder.

3.3.4 With 0.8 plus or minus 0.01 volts from the signal source specifieid in 3.3.1 applied at depth signal terminals C and F, the currents in the "up" coils of both port and starboard circuits shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the "down" coil shall be minimum (1 plus or minus 1 ma).

3.3.4.1 When the follow-up potentiometers specified in 4.4.3.3.1 in the port and starboard circuits are adjusted until the load currents given in 3.3.4 are balanced, both dials on the follow-up potentiometers shall indicate a 10 plus or minus 1 degrees changed.

3.3.5 With 0.8 plus or minus 0.01 volt from the signal source specified in 3.3.1.1 applied at depth signal terminals C and F, the currents in the port "down" coil and the starboard "down" coil shall be maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

(13 plus 1 ma minus 1.5 ma) and the currents in the port "up" coil and the starboard "up" coil shall be minimum (1 plus or minus 1 ma).

3.3.5.1 When the follow-up potentiometers specified in 4.4.3.3.1 in the port and starboard circuits are adjusted until the load currents given in 3.3.5 are balanced, both follow-up potentiometer dials shall indicate a change of 10 plus or minus 1 degrees.

3.3.6 With 0.29 plus or minus 0.01 volt from the signal source specified in 3.3.1 applied at pendulum signal terminals 15 and F, the currents in the port "down" coil and the starboard "up" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the port "up" coil and the starboard "down" coil shall be minimum (1 plus or minus 1 ma).

3.3.6.1 When the follow-up potentiometers specified in 4.4.3.3.1 in the port and starboard circuits are adjusted until the load currents given in 3.3.6 are balanced, both follow-up potentiometer dials shall indicate a change of 10 plus or minus 1 degrees.

3.3.7 With 0.29 plus or minus 0.01 volt from the signal source specified in 3.3.1.1 applied at pendulum signal terminals 15 and F, the currents in the port "up" coil and the starboard "down" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the port "down" coil and the starboard "up" coil shall be minimum (1 plus or minus 1 ma).

3.3.7.1 When the follow-up potentiometers specified in 4.4.3.3.1 in the port and starboard circuits are adjusted until the load currents given in 3.3.7 are balanced, both follow-up potentiometer dials shall indicate a change of 10 plus or minus 1 degrees.

3.3.8 With 0.39 plus or minus 0.01 volt from the signal source specified in 3.3.1 applied at course rate signal terminals 16 and F, the currents in the port "down" coil and the starboard "up" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the port "up" coil and starboard "down" coil shall be minimum (1 plus or minus 1 ma).

3.3.8.1 When the follow-up potentiometers specified in 4.4.3.3.1 on the port and starboard circuits are adjusted until the load currents given in 3.3.8 are balanced, both follow-up potentiometer dials shall indicate a change of 10 plus or minus 1 degrees.

3.3.9 With 0.39 plus or minus 0.01 volt from the signal source specified in 3.3.1.1 applied at course rate signal terminals 16 and F, the currents in the port "up" coil and the starboard "down" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the port "down" coil and the starboard "up" coil shall be minimum (1 plus or minus 1 ma).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

3.3.9.1 When the follow-up potentiometers specified in 4.4.3.3.1 in the port and starboard circuits are adjusted until the load currents given in 3.3.9 are balanced, both follow-up potentiometer dials shall indicate a change of 10 plus or minus 1 degrees.

3.3.10 With 0.33 plus or minus 0.01 volt from the signal source specified in 3.3.1 applied at the course rate signal terminals 16 and F, the current in the rudder "starboard" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "port" coil shall be minimum (1 plus or minus 1 ma).

3.3.10.1 When the follow-up potentiometer specified in 4.4.3.3.1 in the rudder circuit is adjusted until the load currents given in 3.3.10 are balanced, the follow-up potentiometer dial shall indicate a change of 10 plus or minus 1 degrees.

3.3.11 With 0.33 plus or minus 0.01 volt from the signal source specified in 3.3.1.1 applied at the course rate signal terminals 16 and F, the current in the rudder "port" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the current in the "starboard" coil shall be minimum (1 plus or minus 1 ma).

3.3.11.1 When the follow-up potentiometer specified in 4.4.3.3.1 in the rudder circuit is adjusted until the load currents given in 3.3.11 are balanced, the follow-up potentiometer dial shall indicate a change of 10 plus or minus 1 degrees.

3.3.12 With 1.3 plus or minus 0.04 volts from the signal source specified in 3.3.1 applied at the course signal terminals 14 and F, the current in the rudder "starboard" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "port" coil shall be minimum (1 plus or minus 1 ma).

3.3.12.1 When the follow-up potentiometer specified in 4.4.3.1 in the rudder circuit is adjusted until the load currents given in 3.3.12 are balanced, the follow-up potentiometer dial shall indicate a change of 10 plus or minus 1 degrees.

3.3.13 With 1.3 plus or minus 0.04 volts from the signal source specified in 3.3.1.1 applied at the course signal terminals 14 and F, the current in the rudder "port" coil shall be maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "starboard" coil shall be minimum (1 plus or minus 1 ma).

3.3.13.1 When the follow-up potentiometer specified in 4.4.3.3.1 in the rudder circuit is adjusted until the load currents given in 3.3.13 are balanced, the follow-up potentiometer dial shall indicate a change of 10 plus or minus 1 degrees.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

3.3.14 With 1.5 plus or minus 0.01 volts dc applied at any pair of the tachometer input terminals (P and R, S and T, or U and V), the output currents shall increase from balance to at least 10.5 ma in one coil and decrease from balance to a maximum of 3.5 ma in the other coil.

3.3.15 The dc current as drawn from the power source specified in 4.4.3.1.1 shall not exceed 70 ma.

3.3.16 Operating life. The equipment shall be capable of a minimum of 200 hours of operation without servicing, after testing as specified in 4.5.3.

3.3.17 Equipment life. The equipment shall be designed for a minimum shelf life of five years.

3.4 Environmental requirements. The servo amplifier shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.4.1 Vibration. The servo amplifier shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The servo amplifier shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.4.1.2 Vibration, operating. The servo amplifier shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The servo amplifier shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.1.

3.4.2.2 High temperature, operating. The servo amplifier shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.2.

3.4.2.3 Low temperature, nonoperating. The servo amplifier shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.3.

3.4.2.4 Low temperature, operating. The servo amplifier shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. All burrs, sharp edges, and rosin flash shall be removed.

3.6 Workmanship. The servo amplifier, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY CONFORMANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the servo amplifiers shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five servo amplifiers, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the servo amplifier by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five servo amplifiers will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The servo amplifiers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each servo amplifier delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the servo amplifier shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: 65 degrees F to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None.

4.4.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnished and maintain all necessary test equipment and facilities and shall provide personnel for performing the quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

4.4.3.1 Power sources

4.4.3.1.1 DC, 150 volts plus or minus 1 percent at 70 milliamperes.

4.4.3.1.2 AC, 155 volts plus or minus 3 percent, 400 Hertz (Hz) plus or minus 2 percent

4.4.3.1.3 AC, 6.3 volts plus or minus 5 percent, center tapped, 400 Hz plus or minus 2 percent, at 5 amperes.

4.4.3.1.4 A 400 Hz phase reversible, continuously variable signal source of the same phase at the 400-Hz power source described in 4.4.3.1.2 capable of voltage adjustments of 0.01 volt and a maximum voltage of 5 volts. Initial phase adjustment to be such that the voltage on terminals C, 14, 15, and 16 has the same phase as the voltage on terminal 20.

(a) The signal source specified in 4.4.3.1.4 with its phase reversed 180 degrees. That is, the voltage on terminals C, 14, 15 and 16 will have the same phase as the voltage on 19 and the voltage on terminals F and 20 will be in phase.

4.4.3.1.5 Variable 3 volts dc plus or minus 5 percent, 10 ma.

4.4.3.2 Measuring equipment

4.4.3.2.1 Six dc milliammeters capable of reading current values 0 to 15 ma.

4.4.3.2.2 One dc milliammeter capable of reading current values 0 to 100 ma.

4.4.3.2.3 One vacuum tube voltmeter with the following scales: 0 to 0.1, 0.3, 1.0, 3.0 volts ac.

4.4.3.2.4 One dc voltmeter (20,000 ohms per volt) with 0-3 volt scale.

4.4.3.3 Miscellaneous components

4.4.3.3.1 Three potentiometers, hereinafter referred to as port, starboard, and rudder follow-up potentiometers, 10,000 ohms plus or minus 20 percent linear taper, each coupled to a calibrated dial. Terminals are to be marked 1, 2 and 3, with terminal 2 being the swinger arm and terminal 1 the terminal that the swinger approaches when the shaft is rotated clockwise as viewed from the shaft end. Each pot shall be coupled to its dial in such a manner that with 5 volts across the outside terminals (1 and 3), a 1 degree rotation of the dial shall result in a voltage change of 0.1 volt between either outside terminal and the swinger arm. When the dial

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

knob is rotated clockwise, the potentiometer shall also have a clockwise rotation as viewed from the shaft end. The midposition of the swinger arm, halfway between the outside terminals, shall also be the midposition of the dial. This dial midpoint shall be zero degrees, with dial markings progressing upward on each side from this point in steps of one degree, to a maximum of fifteen degrees. Dial calibration shall permit reading the dial position with an accuracy of 0.5 degree.

4.4.3.3.2 One Female Connector, NAVORD Dwg 1263490.

4.4.3.3.3 Six resistors, 5 watts each, 5700 ohms plus or minus 1 percent.

4.4.4 Test procedure

4.4.4.1 Test preparation. Perform the following operations. All references to terminals shall apply to Connector 404, NAVORD Drawing 877408.

4.4.4.1.1 The three 10,000 ohm plus or minus 20 percent linear taper potentiometers specified in 4.4.3.3.1 shall have the following markings:

(a) Two end terminals marked 1 and 3.

(b) Center, variable slider arm terminal marked 2.

4.4.4.1.2 Connect the three follow-up potentiometers, with terminals marked as specified in 4.4.4.1.1 and calibrated as specified in 4.4.3.3.1, to follow-up potentiometer terminals 5 and 6, 7 and 8, and 9 and 10, taking care to connect the potentiometer terminals to the proper leads.

(a) Connect the center slider terminal (2) on the port follow-up potentiometer to terminal D.

(b) Connect the center slider terminal (2) on the starboard follow-up potentiometer to terminal B.

(c) Connect the center slider terminal (2) on the rudder follow-up potentiometer to terminal F.

4.4.4.1.3 Connect the positive terminal of the milliammeter specified in 4.4.3.2.1 to a 5700 ohm resistor (4.4.3.3.3).

(a) Connect the negative terminal of the milliammeter to terminal H.

(b) Connect the other end of the resistor to terminal E.

4.4.4.1.4 Repeat 4.4.4.1.3, connecting five additional milliammeters and five 5700 ohm resistors (4.4.3.3.3) across terminals J, K, L, M, and N, and terminal E (which is common to all connections).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

(a) The current through any of the 5700 ohm resistors and milliammeters specified in 4.4.4.1.4 shall be designated hereafter as the "output current," or "current through the coils."

4.4.4.1.5 Apply power specified in 4.4.3.1.1, 4.4.3.1.2, and 4.4.3.1.3, connected as shown on NAVORD Drawing 877408. Allow a 60-second warmup before continuing.

(a) Center tap of the filament supply specified in 4.4.3.1.3 should be grounded to terminal F.

4.4.4.1.6 Any of the signal input terminals (C, 14, 15 16) not energized during a test may be connected to termanal F.

4.4.4.2 Balancing adjustment. With power applied as specified in 4.4.4.1.5, set the follow-up potentiometer specified in 4.4.3.3.1 at midposition (0 degrees). By rotating zeroing-potentiometers R403, R404, and R405, the output currents from each of the port, starboard, and rudder output circuits can be made equal (henceforth called "balanced") to 6.5 plus or minus 1.5 ma. Repeat the balancing before performing the actions called for in 4.4.4.3.4 through 4.4.4.3.8(a).

4.4.4.3 Sensitivity tests

4.4.4.3.1 All signals shall be considered with respect to ground, and shall be phased as specified in 3.3.1 and 3.3.1.1.

4.4.4.3.2 All test signal voltages shall be 400 Hz plus or minus 2 percent ac voltages.

4.4.4.3.3 Turn the shafts of potentiometers R401 and R402 fully clockwise.

4.4.4.3.4 Using the 400-Hz signal source specified in 4.4.3.1.4, apply 0.8 plus or minus 0.01 volt across depth signal terminals C and F. The currents in the "up" coils of both port and starboard circuits shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the currents in the "down" coils shall decrease to minimum (1 plus or minus 1 ma). Rotate the port follow-up potentiometer clockwise and the starboard follow-up potentiometer counterclockwise (these potentiometers are specified in 4.4.3.3.1) until the output currents balance. At this point the follow-up potentiometer shall have been rotated an amount equivalent to 10 plus or minus 1 degrees on the indicator dial.

(a) Using the signal source specified in 4.4.3.1.4(a), apply 0.8 plus or minus 0.01 volt across depth signal terminals C and F. The currents in the "down" coils of both port and starboard circuits shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

maximum (13 plus 1 ma minus 1.5 ma) and the current in the "up" coils shall be minimum (1 plus or minus 1 ma). Rotate the port follow-up potentiometer counter-clockwise and the starboard follow-up potentiometer clockwise until the output currents balance. At this point the follow-up potentiometer shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

4.4.4.3.5 Using the signal source specified in 4.4.3.1.4, apply 0.29 plus or minus 0.01 volt across pendulum input terminals 15 and F. The currents in the starboard "up" and the port "down" coils shall be maximum (13 plus 1 ma minus 1.5 ma) and the currents in the starboard "down" and the port "up" coils shall be minimum (1 plus or minus 1 ma). Rotate the port and starboard follow-up potentiometers specified in 4.4.3.3.1 counter-clockwise until the output currents balance. At this point the follow-up potentiometers shall have rotated 10 plus or minus 1 degrees, as indicated by the dial.

(a) Using the signal signal specified in 4.4.3.1.4(a), apply 0.29 plus or minus 0.01 volt across pendulum input terminals 15 and F. The currents in the port "up" and the starboard "down" coils shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the currents in the starboard "up" and the port "down" coils shall be minimum (1 plus or minus 1 ma). Rotate the port and starboard follow-up potentiometers specified in 4.4.3.3.1 clockwise until the output currents balance. At this point the follow-up potentiometers shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

4.4.4.3.6 Using the signal source specified in 4.4.3.1.4, apply 0.39 plus or minus 0.01 volt across course rate signal terminals 16 and F. The currents in the port down" coil and the starboard "up" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the currents in the port "up" coil and the starboard "down" coil shall decrease to minimum (1 plus or minus 1 ma). Rotate the port and starboard follow-up potentiometers specified in 4.4.3.3.1 counter-clockwise until the output currents balance. At this point the follow-up potentiometers shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

(a) Using the signal source specified in 4.4.3.1.4(a), apply 0.39 plus or minus 0.01 volt across course rate signal terminals 16 and F. The currents in the port "up" coil and the starboard "down" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the currents in the starboard "up" and the port "down" coils shall be minimum (1 plus or minus 1 ma). Rotate the port and starboard follow-up potentiometers specified in 4.4.3.3.1 clockwise until the output currents balance. At this point the follow-up potentiometers shall have been rotated 10 plus or minus 1 degrees as indicated by the dial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

4.4.4.3.7 Using the signal source specified in 4.4.3.1.4, apply 0.33 plus or minus 0.01 volt across course rate signal terminals 16 and F. The currents in the rudder "starboard" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "port" coil shall decrease to minimum (1 plus or minus 1 ma). Rotate the rudder follow-up potentiometer specified in 4.4.3.3.1 clockwise until the rudder output currents are again balanced. At this point the follow-up potentiometer shall have been rotated 10 plus or minus 1 degrees as indicated by the dial.

(a) Using the signal source specified in 4.4.3.1.4(a), apply 0.33 plus or minus 0.01 volt across course rate signal terminals 16 and F. The current in the rudder "port" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "starboard" coil shall decrease to minimum (1 plus or minus 1 ma). Rotate the rudder follow-up potentiometer specified in 4.4.3.3.1 counterclockwise until the rudder output currents are balanced. At this point the follow-up potentiometer shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

4.4.4.3.8 Using the signal source specified in 4.4.3.1.4, apply 1.3 plus or minus 0.04 volts across course signal terminals 14 and F. The current in the rudder "starboard" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the current in the "port" coil shall decrease to minimum (1 plus or minus 1 ma). Rotate the rudder follow-up potentiometer specified in 4.4.3.3.1 clockwise until the rudder output currents are balanced. At this point the follow-up potentiometer shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

(a) Using the signal source specified in 4.4.3.1.4(a), apply 1.3 plus or minus 0.04 volts across course signal terminals 14 and F. The current in the rudder "port" coil shall increase to maximum (13 plus 1 ma minus 1.5 ma) and the current in the rudder "starboard" coil shall decrease to minimum (1 plus or minus 1 ma). Rotate the rudder follow-up potentiometer specified in 4.4.3.3.1 counterclockwise until the rudder output currents are balanced. At this point the follow-up potentiometer shall have been rotated 10 plus or minus 1 degrees, as indicated by the dial.

4.4.4.4 DC voltage gain test

4.4.4.4.1 With the output currents balanced as specified in 4.4.4.2, apply 1.5 volts plus or minus 0.01 volt dc to terminals P and R, S and T, and U and V in turn, with the positive lead to R, T, and V, respectively.

4.4.4.4.2 The resultant currents shall show an increase (from balance) to at least 10.5 ma in the "down" coils of both port and starboard circuits and the "starboard" coil of the rudder circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

4.4.4.4.3 The currents in the "up" coils of the port and starboard circuits and the "port" coil of the rudder circuit shall show a decrease to 3.5 maximum ma from the balanced state.

4.4.4.5 Total current. Remove the lead from the 150 volts supply to pin E, and insert a 0-100 ma milliammeter (4.4.3.2.2) between the lead and pin E with the negative meter lead to pin F. With 1.3 plus or minus 0.1 volts from the signal source specified in 4.4.3.1.4 applied to C and F, 14 and F, 15 and F, and 16 and F, respectively, the total current indicated on the meter shall be 65 plus 5 minus 10 ma.

4.4.4.6 Visual and dimensional examination. Each servo amplifier shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.7 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The servo amplifier, less pack or packaging, shall be mounted securely to a vibration device by means of its mounting holes and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The servo amplifier shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.5.1.3 Vibration, operating. The servo amplifier shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.5.2 Temperature and humidity

4.5.2.1 High temperature and humidity, nonoperating. The servo-amplifier shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

4.5.2.2 High temperature, operating. The servo amplifier shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.5.2.3 Low temperature, nonoperating. The servo amplifier shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.3.

4.5.2.4 Low temperature, operating. The servo amplifier shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.5.3 Operating life. Apply power sources to the amplifier as specified in 4.4.3.1.1 through 4.4.3.1.4, and allow to operate for 30 minutes. Balance the output currents as specified in 4.4.4.2 and then deenergize the power sources for 30 minutes. Without disturbing the balancing adjustment, cycle the power to the amplifier for 30 minutes on, then 30 minutes off, for a total operating time of 200 hours (400 full cycles).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The servo amplifier shall be packaged in accordance with Method IID of MIL-P-116, LD 162754, and LD 495865.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The servo amplifier shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The servo amplifier shall be packed in accordance with LD 162754 and LD 495865 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

5.2.3 Level B. Not applicable.

5.2.4 Level C. The servo amplifier shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with LD 162754 and LD 495865.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The servo amplifier is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
Navy - OS
Project No. 5990-N048

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:00Z
Check the source to verify that this is the current version before use.